

Class Test 3 (04 September 2026)

1. [50 marks]

Prove that every sequence has a monotone subsequence.

2. [50 marks]

Cauchy Sequence: A sequence of real numbers $\{x_n\}$ is a Cauchy sequence if for any $\epsilon > 0$ there is an integer N such that for all $i, j \geq N$, $|x_i - x_j| < \epsilon$.

Recall that in Homework 3 you have proved the following results:

- Every Cauchy sequence of real numbers is bounded, that is, if $\{x_n\}$ is a Cauchy sequence of real numbers then there exists a real number $B > 0$ such that $|x_n| \leq B$, for all n .
- If a Cauchy sequence $\{x_n\}$ has a subsequence converging to y , then the whole sequence converges to y .

- **Question:** Prove that any Cauchy sequence of real numbers converges.